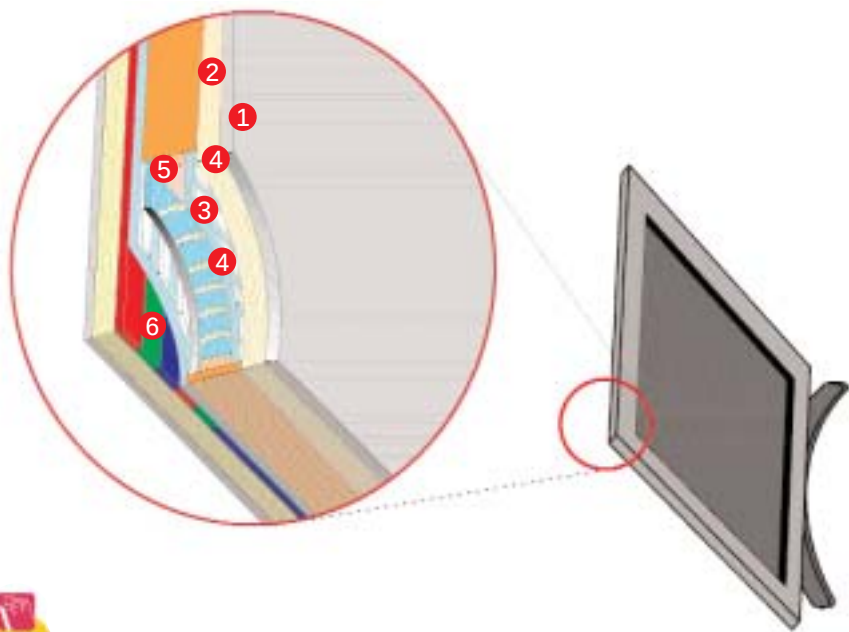




Buying Tips

- ✓ **DIGITAL OR ANALOGUE INPUT:** If you need a sharper display, you can look for a flat panel monitor with a Digital Video Interface (DVI) input. DVI—an emerging standard for digital display connections—transfers data between the PC and monitor in digital format, and so virtually does away with signal loss. Note that you will need a graphics card with a DVI output connector.
- ✓ **RESOLUTION:** Make sure that the native resolution of an LCD monitor is one you are accustomed to. LCD displays are manufactured to work well at a certain resolution, known as the native resolution. While it may be possible to change the resolution of an LCD, setting it to a value other than the native resolution will result in a stretched or blurry visual, or a blank screen. Look for monitors that offer the one-touch auto adjust feature that matches your screen display to your computer's resolution.
- ✓ **DEAD PIXELS:** One thing to look for in LCD monitors is the dead pixel problem. A pixel is a definable location on a monitor screen that consists of multiple, or a single group of phosphor dots (red, green and blue). Dead pixels are those that are always on, or always off. They appear as black or coloured dots on a white screen. A black screen shows them as white or coloured dots. A few dead pixels here and there won't really make a difference, but too many in the wrong place can be an eyesore.

- 1 **Polarizing filter:** This is the filter that polarizes the liquid crystals for display purposes.
- 2 **Glass substrate:** The glass substrate adds the needed reflection and prevents the crystals from being directly exposed.
- 3 **Transparent electrodes:** The transparent electrodes provide the required energy to heat the crystals and change their orientation to present a display.
- 4 **Alignment layer:** The alignment layer takes care of the proper alignment between the electrodes and crystals to give you a perfect display.

Quick Find	
YOU NEED	A space and power saving display solution for home, or office use
Basic Use	
LOOK FOR	A budget 15-inch LCD monitor that provides decent output quality
YOU NEED	A display that reproduces crisp text and images for corporate use
Inter Use	
LOOK FOR	A 17-inch LCD that provides resolutions of up to 1280 x 1024
YOU NEED	A display solution for your media centre PC
Prof Use	
LOOK FOR	An 18-inch LCD monitor that offers high resolution and excellent colour reproduction.

- 5 **Spacer:** It is the layer between the liquid crystals and the transparent electrodes.
- 6 **Colour filter:** This is responsible for imparting colours to the crystals.

Facts

- The rate of failure increases with the size of the LCD screen. This is one of the primary reasons for LCD monitors being so expensive.
- The power consumption of an LCD is around one-sixth that of a CRT monitor.
- The screen size of an LCD is exactly as described—i.e., 17-inch is 17 inches, and not 15.3 inches as in a CRT. So, an LCD screen is actually equivalent to the next standard size of a CRT. For example, a 17-inch LCD would actually be equivalent to a 19-inch CRT.



The Future

LCD screens have intruded into the CRT segment, and are gaining wider acceptance in the market day by day. However, there are a couple of new technologies on the horizon waiting for a push. One such technology is Plasma display, which hasn't really made a mark because of the very high costs involved. Plasma displays are toys of the rich and famous and do their bit well for home theatres, rather than for business purposes. On the other hand is OLED—a technology that's already made its presence in the small screens of digital cameras, but has yet to make an impact on the PC front. Just like other new technologies that have already been developed, but are waiting for mass acceptance, OLED will also need large quantities of key material and fabrication plants. The process will take time, lots of convincing, and much interaction with vendors.

Apart these, there are several others in the making, such as the Ferro Electric Display, and Digital Light Processing—Micro Mirror, Direct/Digital Image Light Amplifier—that hold a lot of promise.